IHTC Model Transformations

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MOVES / MODEL TRANSFORMATIONS

Decisions

D1: What day in the planning horizon will each patient be admitted?

D2: What operating theatre will each patient use?

D3: What recovery room will each patient use?

D4: What rooms will each nurse cover on their assigned shifts?

Model Transformations

0.1 Patient Admission and Removal

- **T1.**
  Type: atomic.
  Description: Insert a random non-mandatory patient with a random admission day, room, and theatre.
  Classes used: `Patient`, `Admission`, `Room`, `OperatingTheatre`.
  Purpose: to increase the number of scheduled patients and explore new possible allocations quickly.
- **T2.**
  Type: atomic.
  Description: Insert the non-mandatory patient with the least workload into a feasible admission day, room, and theatre.
  Classes used: `Patient`, `Admission`, `Room`, `OperatingTheatre`, `SurgeonAvailability`, `OperatingTheatreAvailability`, `RoomAvailability`.
  Purpose: to add a patient who is easier to fit into the schedule.
- **T3.**
  Type: atomic.
  Description: Find the earliest available surgeon, then find a theatre and room available on the same day, and insert a random compatible non-mandatory patient.
  Classes used: `Patient`, `Admission`, `Surgeon`, `SurgeonAvailability`, `OperatingTheatre`, `OperatingTheatreAvailability`, `Room`, `RoomAvailability`.
  Purpose: to insert a patient in a way that is more strongly guided by actual resource availability, especially surgeon and theatre constraints.
- **T4.**
  Type: atomic.
  Description: Remove a random non-mandatory patient.
  Classes used: `Patient`, `Admission`.
  Purpose: to free up resources while only sacrificing an optional patient, which helps repair overloaded or poor-quality schedules.

- **T5.**
  Type: atomic.
  Description: Remove a random patient.
  Classes used: `Patient`, `Admission`.
  Purpose: to make a larger change to the schedule and free capacity, even if that means removing a more important patient.
- **T6.**
  Type: joint (T4 + T1).
  Description: Remove a random non-mandatory patient, then insert a random non-mandatory patient.
  Classes used: `Patient`, `Admission`, `Room`, `OperatingTheatre`, `SurgeonAvailability`, `OperatingTheatreAvailability`, `RoomAvailability`.
  Purpose: to swap one optional patient for another, testing whether a different patient gives a better schedule without changing the total number of optional patients.
- **T7.**
  Type: joint (T5 + T1).
  Description: Remove a random patient, then insert a random non-mandatory patient.
  Classes used: `Patient`, `Admission`, `Room`, `OperatingTheatre`, `SurgeonAvailability`, `OperatingTheatreAvailability`, `RoomAvailability`.
  Purpose: to replace part of the current schedule with a new optional patient, allowing a more disruptive change in the set of admitted patients.

0.2 Patient Reassignment

- **T8.**
  Type: atomic.
  Description: Give a random admitted patient a random new compatible room.
  Classes used: `Patient`, `Admission`, `Room`, `RoomAvailability`.
  Purpose: to reassign a patient to a different feasible room in order to improve room usage or reduce incompatibilities.
- **T9.**
  Type: atomic.
  Description: Give a random admitted patient a random new compatible admission day.
  Classes used: `Patient`, `Admission`, `SurgeonAvailability`, `OperatingTheatreAvailability`, `RoomAvailability`.
  Purpose: to move a patient to a different feasible day in order to improve schedule balance or resource usage.
- **T10.**
  Type: atomic.
  Description: Give a random admitted patient a random new compatible theatre.
  Classes used: `Patient`, `Admission`, `OperatingTheatre`, `OperatingTheatreAvailability`.
  Purpose: to reassign a patient to a different feasible theatre in order to improve theatre allocation.
- **T11.**
  Type: joint (T8 + T9 + T10).
  Description: Change a patient's room, admission day, and theatre sequentially.
  Classes used: `Patient`, `Admission`, `Room`, `RoomAvailability`, `OperatingTheatre`, `OperatingTheatreAvailability`, `SurgeonAvailability`.
  Purpose: to make a broader coordinated change to one patient's allocation across several decision variables.
- **T12.**
  Type: joint (T8 + T9).
  Description: Change a patient's room and admission day sequentially.
  Classes used: `Patient`, `Admission`, `Room`, `RoomAvailability`, `SurgeonAvailability`, `OperatingTheatreAvailability`.
  Purpose: to jointly adjust accommodation and timing in a way that may improve feasibility or quality more than changing one variable alone.
- **T13.**
  Type: joint (T9 + T10).
  Description: Change a patient's admission day and theatre sequentially.

Classes used: `Patient`, `Admission`, `OperatingTheatre`, `OperatingTheatreAvailability`, `SurgeonAvailability`, `RoomAvailability`.
Purpose: to jointly modify the timing and surgical allocation of a patient while keeping the room unchanged.

0.3 Nurse Assignment

- **T14.**
  Type: atomic.
  Description: Add a random nurse to a random compatible room on a random compatible shift.
  Classes used: `Nurse`, `NurseWorkingShift`, `HospitalisationShift`, `Room`, `RoomShiftAssignment`.
  Purpose: to create a new nurse-to-room assignment where the nurse is available on that shift and satisfies the required skill and workload conditions.
- **T15.**
  Type: atomic.
  Description: Remove a random nurse from a random room on their schedule.
  Classes used: `Nurse`, `HospitalisationShift`, `RoomShiftAssignment`.
  Purpose: to remove an existing nurse-to-room assignment and allow the optimisation to repair or rebalance staffing decisions.
- **T16.**
  Type: joint (T15 + T14).
  Description: Remove a nurse from one room and add them to another.
  Classes used: `Nurse`, `NurseWorkingShift`, `HospitalisationShift`, `Room`, `RoomShiftAssignment`.
  Purpose: to reallocate nursing capacity between room-shifts without changing the total number of nurse assignments.